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Hybrid Data-Driven Modeling and Prediction of Photovoltaic Soiling Losses: Balancing Accuracy and Simplicity

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ABSTRACT

Photovoltaic (PV) system efficiency is significantly affected by soiling, leading to energy losses and increased operational costs. Within the SERENDI-PV and CACTUS projects, this study refines soiling loss modeling through two complementary approaches: the stochastic quantifying soiling loss (SQL) method and a machine learning (ML)-based predictive model. SQL quantifies soiling losses using electrical performance data, independent of meteorological inputs, while the ML model forecasts losses using environmental parameters such as temperature, humidity, wind speed, and particulate matter concentration. The SQL method identifies soiling phases—cleaning periods, stable periods, and soiling periods—by analyzing PV performance trends. Monte Carlo simulations generate probable soiling profiles, assessing uncertainty and informing maintenance strategies. Additionally, SQL enables classification of soiling accumulation patterns, distinguishing between gradual and rapid soiling events. The ML-based approach integrates artificial neural networks and regression models to predict soiling losses, applying data preprocessing techniques to enhance accuracy. Regional climate variations and site-specific soiling characteristics are incorporated to improve predictive performance. Preliminary results validate the robustness of both methodologies. SQL-generated profiles align closely with observed soiling trends, and ML models effectively capture seasonal variations. Field studies at a utility-scale PV plant in Spain further demonstrate the location dependency of soiling patterns. Comparative analysis of SQL predictions and real soiling measurements shows prediction errors between 1% and 2.6% over an 8-day forecast period. These findings support the integration of predictive soiling models into PV monitoring platforms, optimizing maintenance strategies, and minimizing energy yield losses.

1 | Introduction: State-of-Play and Rationale/Objective

Photovoltaic (PV) systems are significantly impacted by soiling, a phenomenon that reduces their efficiency due to the accumulation of dust and other particulate matter (PM) on the surface of PV modules. This leads to energy losses and increased

operational costs, making it a critical factor in PV performance optimization [1–3]. The review study in [1] details the evolution of soiling research, assessment methods, and mitigation techniques like cleaning and antisoiling coatings; whereas the review in [2] highlights the severity of soiling losses in arid and polluted areas, as well as the advantages of passive antisoiling coatings over active cleaning for their waterless operation, lower

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cost, and durability. In [3], Bess et al. underscore the need for site-specific soiling monitoring, evaluating various technologies, and projecting a growing global demand for tens of thousands of sensors annually. The authors also draw experimental comparisons and analyze the economic viability of soiling mitigation and monitoring.

The economic repercussions of soiling in PV systems are substantial, with global income losses estimated to even exceed 10 billion EUR annually due to soiling-induced performance degradation [4]. Soiling severity depends on multiple factors, including local climate (e.g., rainfall frequency, wind speed, humidity), airborne PM composition (PM₁₀, PM_{2.5}), and site-specific conditions (e.g., proximity to deserts, agriculture, or urban pollution). Soiling loss estimates vary widely depending on geographical location, environmental conditions, and cleaning frequency. Therefore, the accurate quantification, modeling, and prediction of soiling losses are crucial for optimizing cleaning schedules, improving energy yield forecasts, and facilitating proactive maintenance strategies [5–7]. More particular, IEA PVPS Task 13 experts in both [5] and [6] summarized soiling from multiple perspectives, including dust particle types and global distribution, mechanisms, soiling sensors and measurement techniques, modeling approaches, economic impacts, mitigation strategies, including climate-specific preventive maintenance for soiling prone PV plants, as well as considerations for snow shading as PV systems expand into higher latitudes. In [7] researchers developed two novel soiling testing prototypes offering accurate, calibration-free power loss estimates. Their lab protocol also revealed new insights into soiling mechanisms and site-specific effects.

Despite the critical impact of soiling, its accurate assessment remains a complex task. Several models have been recently developed and validated using data from specific systems or locations, which raises fair concerns about their generalizability to diverse site- and climate-specific environments. The dynamic and complex nature of soiling, influenced by multiple individual or combined environmental factors, necessitates robust modeling techniques that can capture these intricate relationships [8, 9].

The approaches to soiling modeling can be broadly categorized into analytical and physics-based models, empirical and statistical models, and data-driven models. Analytical models use mathematical relationships derived from environmental data, such as rainfall, wind, and PM concentration, to estimate soiling losses. Physics-based models, a more detailed subcategory, delve into the fundamental mechanisms of dust deposition, adhesion, and removal, incorporating factors like particle size, wind speed, relative humidity, and precipitation. These models offer a detailed understanding of the soiling phenomenon by simulating the physical interactions at the module surface.

One of the earliest and simplest analytical approaches is the Kimber model [10], which assumes linear soiling accumulation during dry periods and complete cleaning when rainfall exceeds a fixed threshold. While straightforward, it does not account for PM concentration or partial cleaning effects. More advanced models have since been developed to address these limitations. The Humboldt State University (HSU) model [11] incorporates PM concentration (PM₁₀ and PM_{2.5}), deposition velocity, and tilt angle for a more nuanced approach. Toth et al. [12]

proposed a model that differentiates between fine and coarse particles, capturing the adhesive properties of fine particles more effectively.

A comparative analysis of these models using data from Spanish PV plants [13] showed that the Kimber model was the simplest but least accurate. The HSU and Toth models improved accuracy by incorporating PM but were still limited by fixed cleaning thresholds. A particularly adaptable model, SOMOSclean [13], employs an exponential growth function to simulate soiling saturation and incorporates variable cleaning thresholds based on rainfall intensity. It demonstrated the best performance, with a mean absolute error of 0.694%, due to its dynamic handling of rainfall and dust events. A primary limitation of physics-based models, however, is their demanding need for extensive, site-specific environmental data and precise characterization of dust properties, which can be logistically challenging.

Empirical and statistical models leverage historical PV performance data to quantify soiling losses. A common approach is to use the soiling ratio (SR), which compares the performance of a soiled module to that of a clean reference module [7]. This allows for the direct calculation of soiling losses. Other mathematical models and statistical methods calculate a so-called dirt derating factor [14] or analyze performance trends to infer soiling rates [15, 16]. While these approaches are straightforward, their accuracy can be compromised by the availability of a truly clean reference or the precise identification of clean periods.

The increasing availability of large-scale PV performance and environmental datasets has propelled the development of machine learning (ML) models for soiling prediction. These data-driven approaches are adept at capturing complex, nonlinear relationships that might be intractable for traditional models. Common ML methods include linear regression and artificial neural networks (ANNs), with ANNs often demonstrating superior results. For example, Laarabi et al. [17] and Chiteka et al. [18] applied ANNs to predict soiling losses using environmental variables like rainfall, humidity, and PM, showing improved accuracy but being limited by data availability and computational complexity. Zitouni et al. [19] combined ML with regression techniques, highlighting the potential of hybrid methods. A notable data-driven approach is the stochastic rate and recovery (SRR) method, which automatically detects soiling and cleaning events and models soiling rates from a single system's performance data, without needing a separate clean ref. [15].

Studies emphasize the importance of using hourly data to account for intraday variations in environmental factors and training models over extended periods to capture seasonal patterns. Key environmental inputs often include ambient temperature, relative humidity, wind speed and direction, rainfall, and airborne PM, along with system properties like tilt angle, in order to extract “PV soiling profiles” for analysis, forecasting and optimized cleaning [20]. Data processing involves standardization, cleaning, and filtering for specific conditions like snowy days. A recognized limitation of many ML models is that they are trained and tested on data from the same system or location, thus highlighting the need for multienvironment training to enhance generalizability. Future work in this domain includes exploring advanced neural network architectures (e.g., time-series neural

networks, recurrent neural networks) and integrating specific considerations for bifacial PV systems.

Finally, in the critical fields of soiling characterization/analysis and forecasting, recent research has yielded significant improvements toward advanced modeling and assessment of PV soiling losses. Micheli et al. [20] developed a framework for generating site-specific soiling profiles, demonstrating that median-based annualized soiling rates often fail to capture seasonal variations. Their weighted average approach and Markov-chain weather generation algorithms achieved prediction errors <1.3% for average soiling and could identify optimal cleaning dates within 3 weeks of actual needs, representing a substantial improvement over conventional methods. Kumar et al. [21] advanced empirical modeling through modifications of established frameworks, testing five variants of the Kimber and HSU models against optical sensor data. Their optimized models reduced root mean square error by up to 23% compared to baseline versions, with all configurations maintaining mean absolute percentage error below 1%. Notably, they identified distinct model performance characteristics: modified Kimber models excelled during extended dry periods (model 1) and frequent rainfall conditions (model 3), while their adapted HSU formulation showed 12% greater accuracy than the original. These findings emphasize the importance of site-specific model selection based on local precipitation patterns. Alternative approaches to soiling prediction have also emerged. Ballestrín et al. [22] demonstrated the effectiveness of autoregressive moving-average time-series models, achieving mean relative errors of just 0.07% for seasonal soiling forecasts in PV systems. However, their analysis revealed limitations in predicting daily fluctuations, suggesting the need for higher-resolution data inputs. Smestad et al. [23] and Micheli et al. [24] pioneered spectral characterization and image analysis, respectively. In [23], researchers established that naturally deposited soiling follows consistent transmittance patterns described by a modified Ångström turbidity equation. Their work also provided a valuable cross-validation framework by correlating cleanliness standards with optical measurements ($R^2 = 0.89$). Meanwhile, researchers in [24] investigated the uncertainties of various image analysis thresholding methods. Their research, based on a large dataset of micrographs, found some methods like “percentile” and bimodal distributions to be inconsistent. They identified the “Triangle” method as the most reliable and recommended taking more than one micrograph per sample to account for nonuniform soiling and keep the estimation error below a specified threshold. The convergence of optical [23, 24], statistical [22], and ML enhanced [21] approaches points toward an emerging paradigm of multimethod soiling assessment.

Despite advancements, soiling modeling faces several challenges, including the incomplete integration of environmental variables like wind and humidity, and the limitations of sensors. ML models, while powerful, require large datasets, which can limit their applicability in regions with sparse monitoring. Within the SERENDI-PV project, this study aims to refine soiling loss modeling and develop predictive models to enhance PV plant maintenance and energy yield forecasting. This work presents two complementary approaches: the stochastic quantifying soiling loss (SQL) method, which provides retrospective analysis and quantification of soiling impact from electrical performance data alone; and a ML-based approach that aims at forward-looking predictions

using external environmental data. Our ultimate goal is to integrate these two models into predictive tools for improved PV energy yield assessments, ensuring that operators can make data-driven decisions to minimize energy yield losses due to soiling.

2 | Methodology and Models Architecture

2.1 | SQL Method Component

The SQL method identifies soiling trends by analyzing PV performance metric(s) (PerfM) over time. In the context of this study, PerfM refers to the normalized ratio between measured electrical variables (e.g., I_{DC} , P_{DC} , PR_{DC} , P_{AC}) and their theoretical clean-standard test conditions (STC) values. This PerfM serves as a proxy to derive the SR in the absence of a clean reference module. Unlike traditional methods that require external environmental inputs such as rainfall data, SQL relies entirely on variations in electrical performance, making it particularly useful in cases where meteorological data is unavailable or unreliable. In such approach, statistical thresholds are defined to segment performance variations into distinct “key” phases (periods) that make up the soiling profile:

2.1.1 | Cleaning Day (CD)

Defined as a day when the subsequent day’s PerfM significantly increases, specifically if it exceeds that day’s PerfM by a threshold. This threshold is calculated as the third quartile plus 1.5 times the interquartile range of all daily PerfM variations.

2.1.2 | Stable Period (St.P)

Characterized by days where the PerfM remains consistently close to the previous day’s value, meaning the difference is below the aforementioned threshold.

2.1.3 | Stable Evolution Period

A sub-period within the St.P, identified when the PerfM difference from the first day of the period exceeds the threshold.

2.1.4 | Soiling Period (SP)

Comprises all other days not classified as CD or part of St.Ps.

Cleaning events are identified based on a significant increase in PV PerfMs, while St.Ps are detected by analyzing fluctuations within a defined threshold. The transition into a SP is determined once PV performance metrics imply degradation beyond acceptable variation limits. A theoretical representation of these different phases is given in Figure 1, illustrating the transitions between cleaning, St.P, and SPs. To contextualize such phases/classifications into real-life profiles of soiling losses, Figure 2 shows an example profile from actual real-case data, displaying how these phases are detected in practice.

Following the characterization of each period family, we employ statistical analyses and Monte Carlo simulations, to generate probable soiling profiles, enabling estimates of average SRs and uncertainty ranges. This probabilistic approach helps define

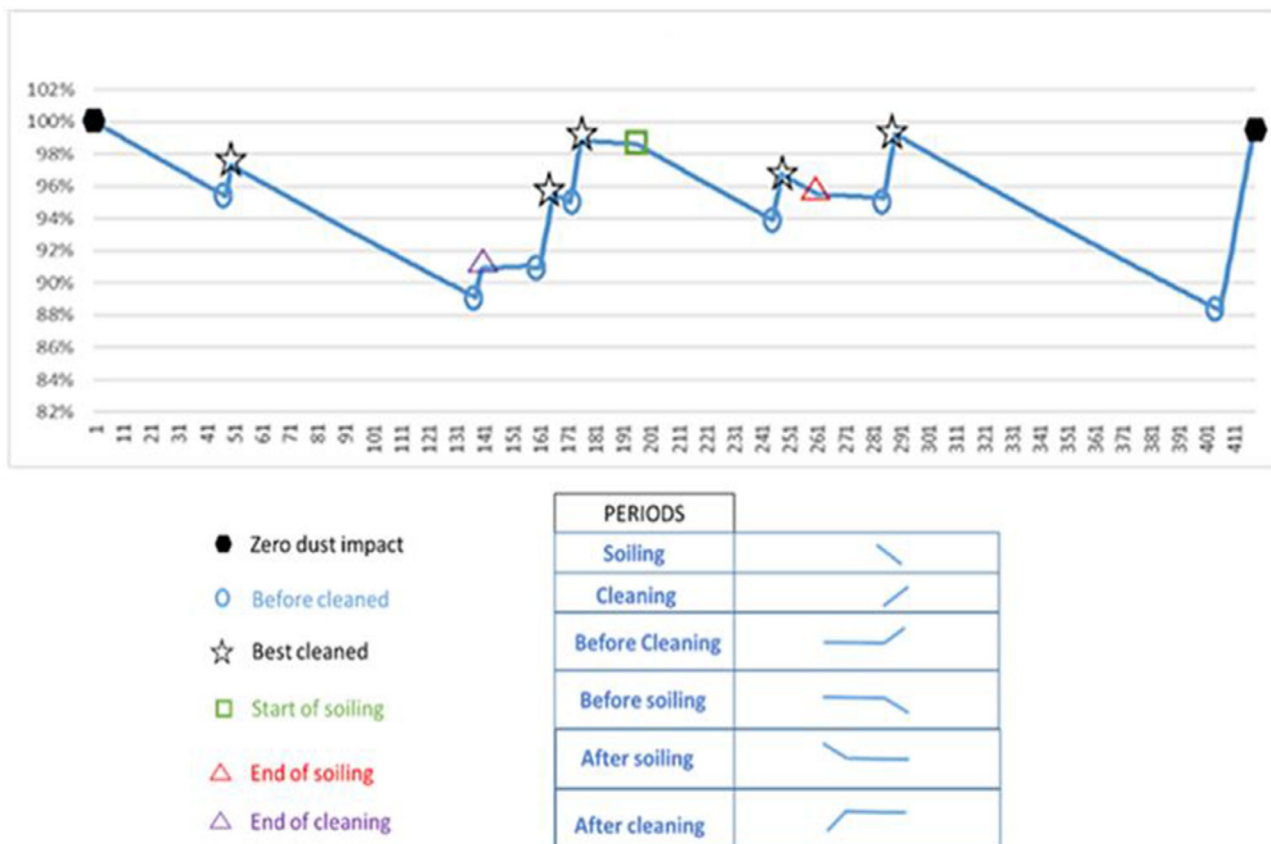


FIGURE 1 | Theoretical representation of the different key phases-variations in PV performance, considered for soiling profiling in the SSQL approach.

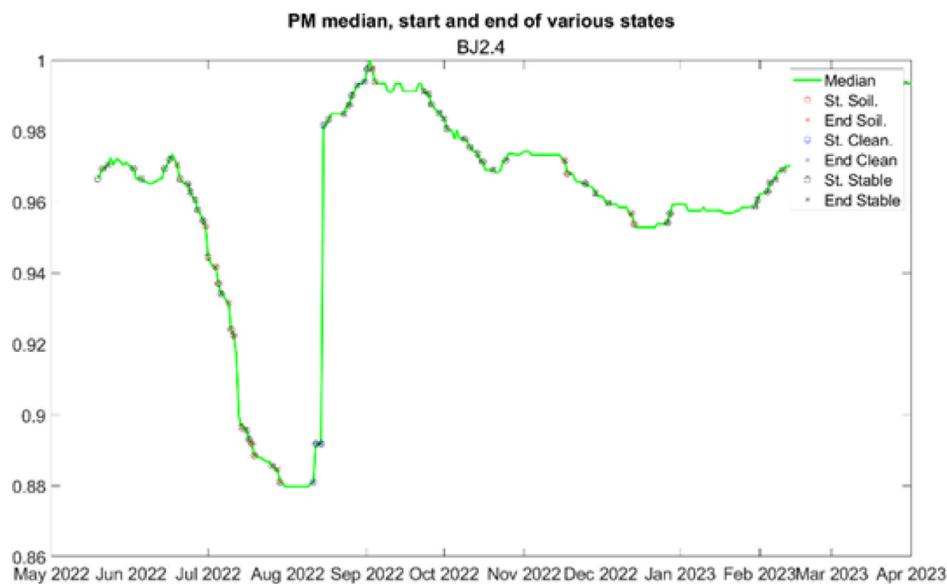


FIGURE 2 | Identification of such phases within a real-case performance/soiling data from a utility-scale PV plant.

different soiling scenarios, ranging from worst-case to optimal conditions. The methodology also includes sensitivity analysis, examining how variations in input parameters, such as the frequency and effectiveness of cleaning events, influence the estimated soiling losses. Additionally, the SSQL method allows the classification of different soiling accumulation patterns by analyzing the rate of PV performance decline during SPs.

This enables distinguishing between slow, gradual soiling accumulation and rapid degradation due to specific environmental conditions. Such “soiling profiling” capability provides valuable insights for optimizing maintenance strategies for PV plants.

It is important to note that the current SSQL approach does not account for potential seasonal influences or correlations between

parameters like cleaning efficiency and PM levels. Seasonal effects such as rainfall cycles, agricultural activity, wind-driven dust transport, or humidity can strongly affect deposition and natural cleaning processes. By not explicitly accounting for these cycles, SQSL may underestimate or overestimate long-term soiling trends in regions with pronounced seasonal patterns. This limitation arises because SQSL was deliberately designed as a stand-alone performance-based tool for cases where reliable meteorological data are unavailable.

The effectiveness of the SQSL method hinges on the accurate measurement and reliability of the chosen electrical variable for calculating the PerfM. The selected variable must meet three crucial conditions: high measurement accuracy, reliable data transmission, and clear identifiability of discrepancies due to soiling or cleaning. Based on these criteria, the following variables have been identified in order of relevance from most to least:

1. DC current (I_{DC}) at the inverter or junction box input
2. DC power (P_{DC}) at the inverter or junction box input
3. Performance ratio calculated with DC power (PR_{DC})
4. AC power (P_{AC}) at the inverter output

The choice and order of these variables are motivated by their respective sensitivity to soiling and robustness against confounding factors. I_{DC} is generally the most direct indicator of irradiance-to-current conversion and thus most sensitive to optical transmission losses caused by soiling. P_{DC} integrates both current and voltage, providing a broader performance measure, but with slightly lower sensitivity to soiling compared to I_{DC} . PR_{DC} is useful for normalization across sites, although it is subject to uncertainties from irradiance sensor calibration and reference conditions. Finally, P_{AC} reflects overall energy yield but is more strongly influenced by inverter efficiency variations and potential curtailment events, which reduce its diagnostic value for soiling.

Other electrical variables could, in principle, also be used. Short-circuit current (I_{SC}) is highly sensitive to optical losses and often

used in dedicated soiling stations, but it is rarely monitored continuously at utility scale. Open-circuit voltage (V_{OC}) is less sensitive to soiling and more strongly affected by temperature, making it less reliable in this context. Furthermore, advanced monitoring could integrate spectral mismatch indicators or module-level maximum power point tracking data, though such variables are not typically available in standard plant supervisory control and data acquisition systems. For these reasons, the selected variables (I_{DC} , P_{DC} , PR_{DC} , P_{AC}) represent a balance between sensitivity to soiling and availability in operational PV plants.

All measurements are corrected for temperature and irradiance to standardize them to STC (25°C, 1000 W/m² AM1.5). The PerfM is then defined as the ratio between the corrected measurement and the theoretical value of the same variable under STC with a perfectly clean PV module.

2.2 | ML Component

To enhance CEA's diagnostic/predictive framework for PV soiling losses, the above SQSL method is coupled and complemented with an ML-based methodology, integrating ANNs and regression modeling to predict soiling losses from environmental parameters. This approach aims to address limitations of existing soiling models, which often lack generalization across diverse environments, by incorporating more parameters, utilizing shorter sampling periods (hourly data), and training over extended durations to capture seasonal variations. The overall architecture of the algorithm for data processing and soiling loss modeling is illustrated in Figure 3.

The proposed model integrates linear regression and ANNs, utilizing hourly data to capture intraday variations in environmental factors. Input parameters are categorized into three groups based on literature review and their demonstrated influence: Environmental factors, including ambient temperature, relative humidity, wind (mean, peak speed, and direction), rainfall (accumulation and intensity), particle matter (<10 µm, airborne density), and horizontal albedo (snow data is used solely for filtering snowy days); system properties, which are constant values characterizing the system, such as tilt angle, module orientation, coating, geographical

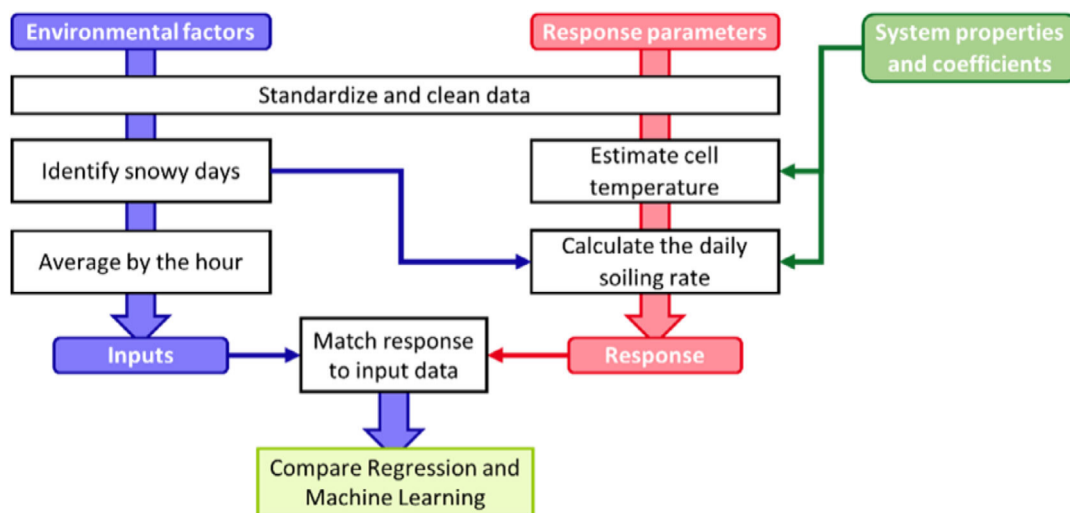


FIGURE 3 | Architecture of the ML-based model for soiling loss prediction, assessed complementarily in this study.

coordinates (latitude, longitude, altitude), PV module temperature coefficient, and IV curves at STC for at least PV module; and response parameters, comprising electrical measurements and parameters used for model response calculation, including current, power, energy, in-plane irradiance, and PV module temperature.

The data processing workflow involves several critical steps:

1. Standardization and cleaning: Environmental and response parameters undergo initial standardization and cleaning, removing physically impossible values. Missing precipitation data can be interpolated.
2. Snowy day identification: For snowy regions, days affected by snow are identified and excluded from SR calculations. An effective filter leverages the strong correlation between increased albedo and snow presence to address noisy sensor data.
3. Hourly averaging: Data are averaged by the hour to capture daily environmental variations. Specific aggregations apply: maximum for peak wind speed, sum for rainfall, and average

of positive values only for rain intensity. These hourly averages serve as input for regression and ML models.

4. Soiling rate calculation: The daily soiling rate, representing the daily variation of the SR, is calculated from response parameters (Figure 3). This calculation involves estimating cell temperature (if not measured, using Sandia's or nominal operating cell temperature models), correcting the PerfM (preferably DC current or power, measured at the inverter's DC input, smoothed with a 1 h moving average), calculating a temperature-corrected instantaneous PerfM using irradiance, cell temperature, and the metric with synchronized data, and filtering for irradiance exceeding 500 W/m^2 to minimize noise. For non-snowy days, the irradiance-weighted daily average of the performance ratio is computed; this daily PerfM is then normalized to the 98th percentile (with a 5% tolerance) to derive the daily SR, further refined using a 28-day moving median (± 14 days).

The processed hourly environmental parameters and the daily soiling rate form a comprehensive dataset for training and comparing various regression and ML models.

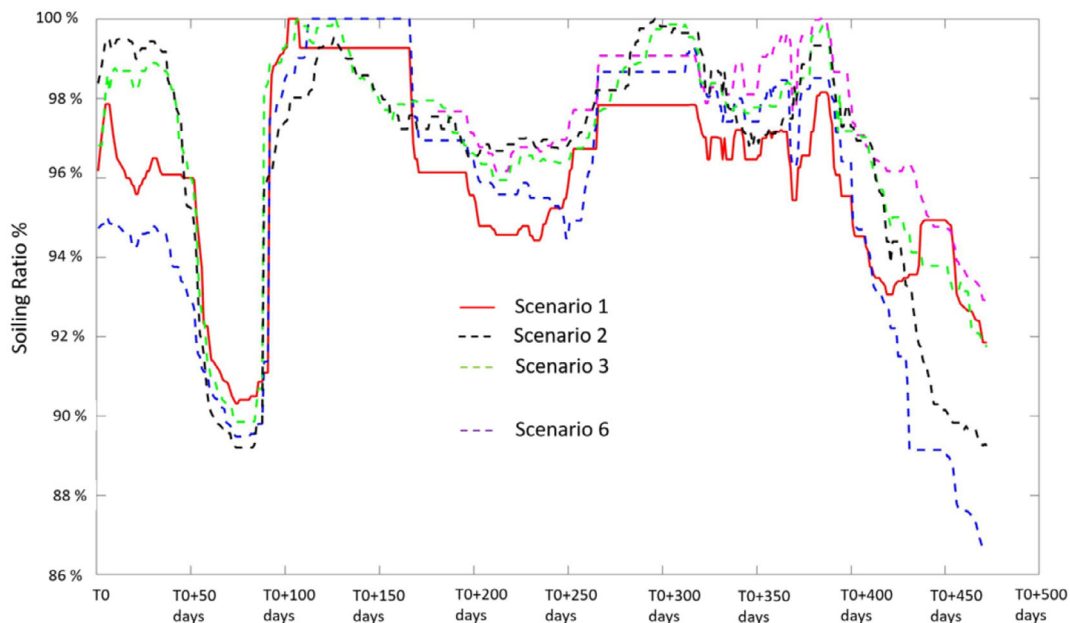


FIGURE 4 | Example results of SQL-based generation of various soiling loss profiles, based on the real-case data/curve shown in Figure 1.

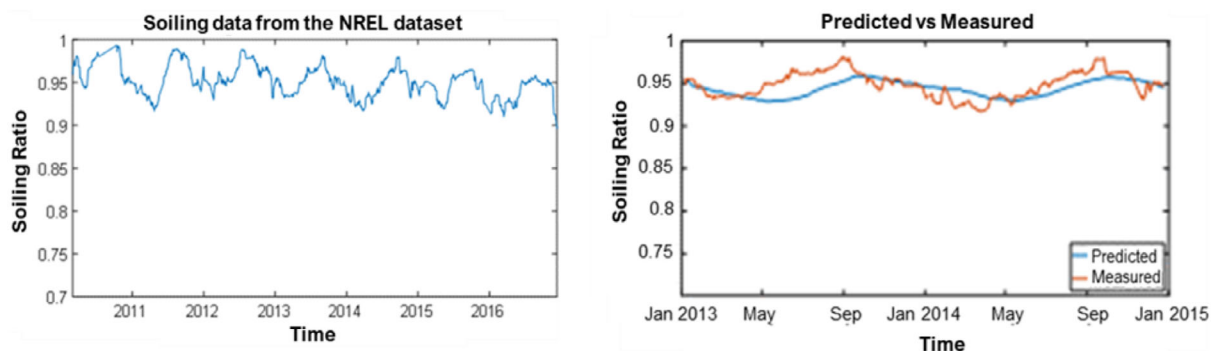


FIGURE 5 | Left: Measured SR for the case of NREL dataset, in a 7-year period (01/2010-01/2017). Right: Measured versus predicted (through our ML-based model) SR, over 2 years (01/2013-01/2015) of the same NREL dataset.

3 | Validation – Results and Discussion

First SQL validation results indicate a high correlation between simulated and actual soiling profiles. The Monte Carlo-generated profiles effectively reconstruct historical soiling trends, confirming the robustness of the method. Figure 4 demonstrates this through a set of simulated PV performance (soiling loss) profiles, for the real-case study data of Figure 1-right, illustrating how uncertainty ranges are captured in the estimation. The postcleaning PV performance jumps observed in field data validate the

method's ability to estimate cleaning efficiencies under different operational conditions.

Further, the ML-based models trained on datasets from NREL and CEA floating PV plants demonstrate a strong ability to capture seasonal soiling variations (Figure 5 and 6). The comparison of measured and predicted SRs reveals that prolonged training periods improve predictive accuracy. However, the models require further refinement to account for location-specific environmental influences.

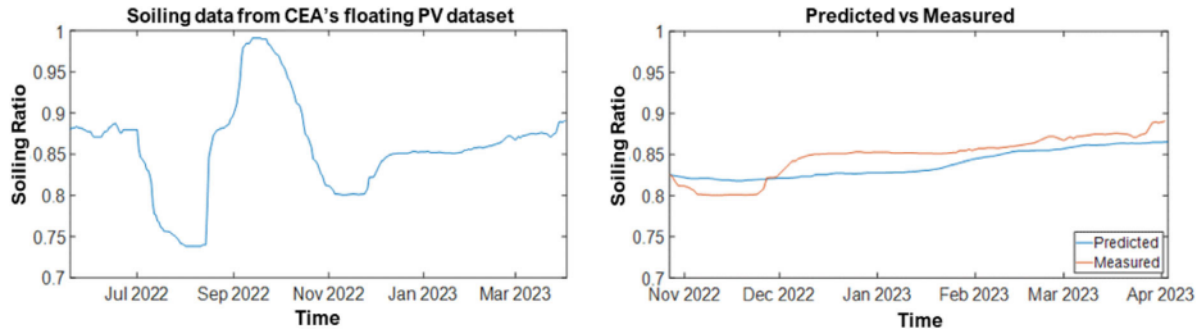


FIGURE 6 | Left: Measured SR for the case of CEA's dataset from a commercial floating PV plant. Right: Measured versus predicted (through the CEA ML-based model) SR, over the same period.

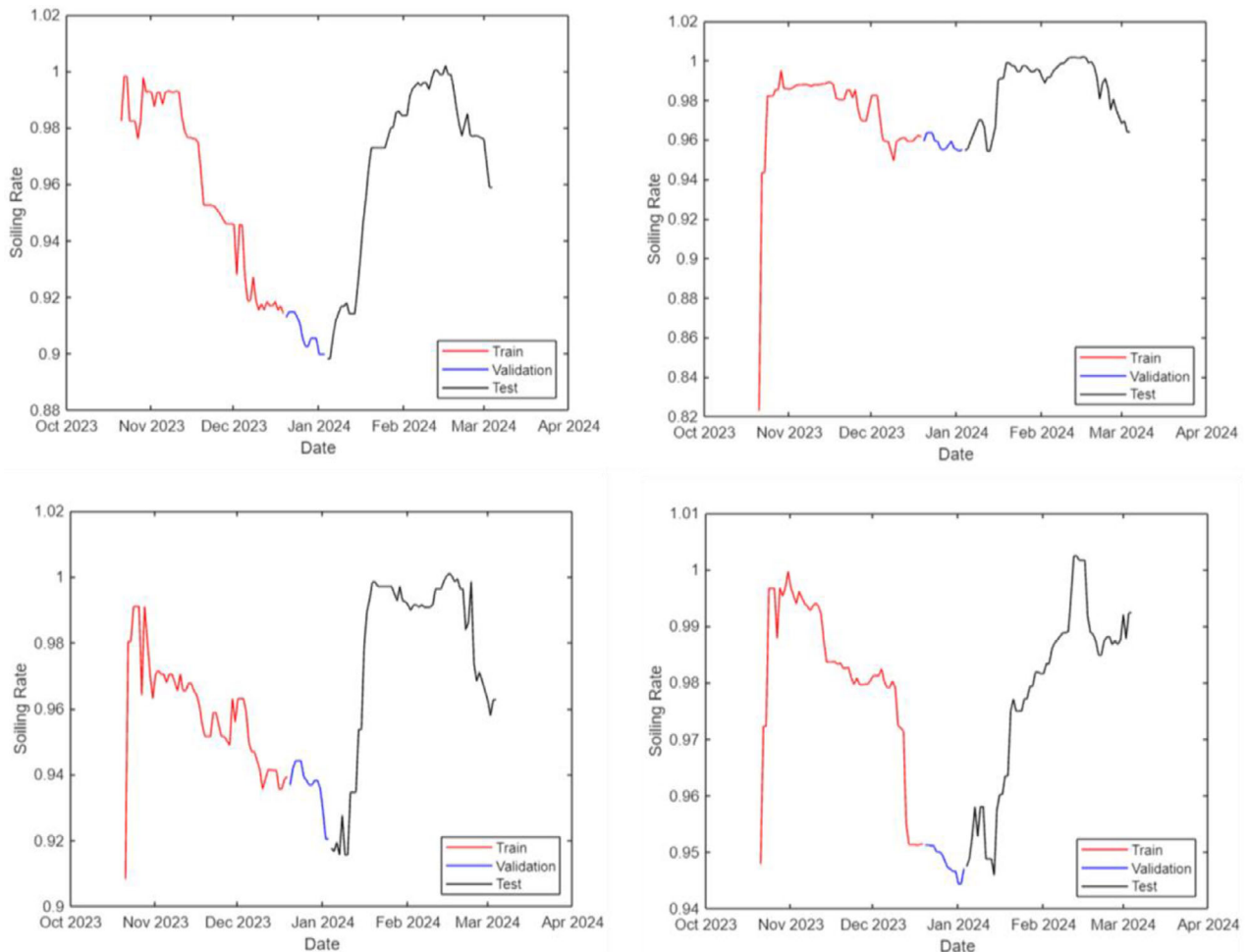


FIGURE 7 | Soiling rate trends for different PV strings (four indicative examples) of the studied utility-scale PV plant in northeastern Spain. Labels “train,” “validation,” and “test” indicate the data split used for model training and evaluation.

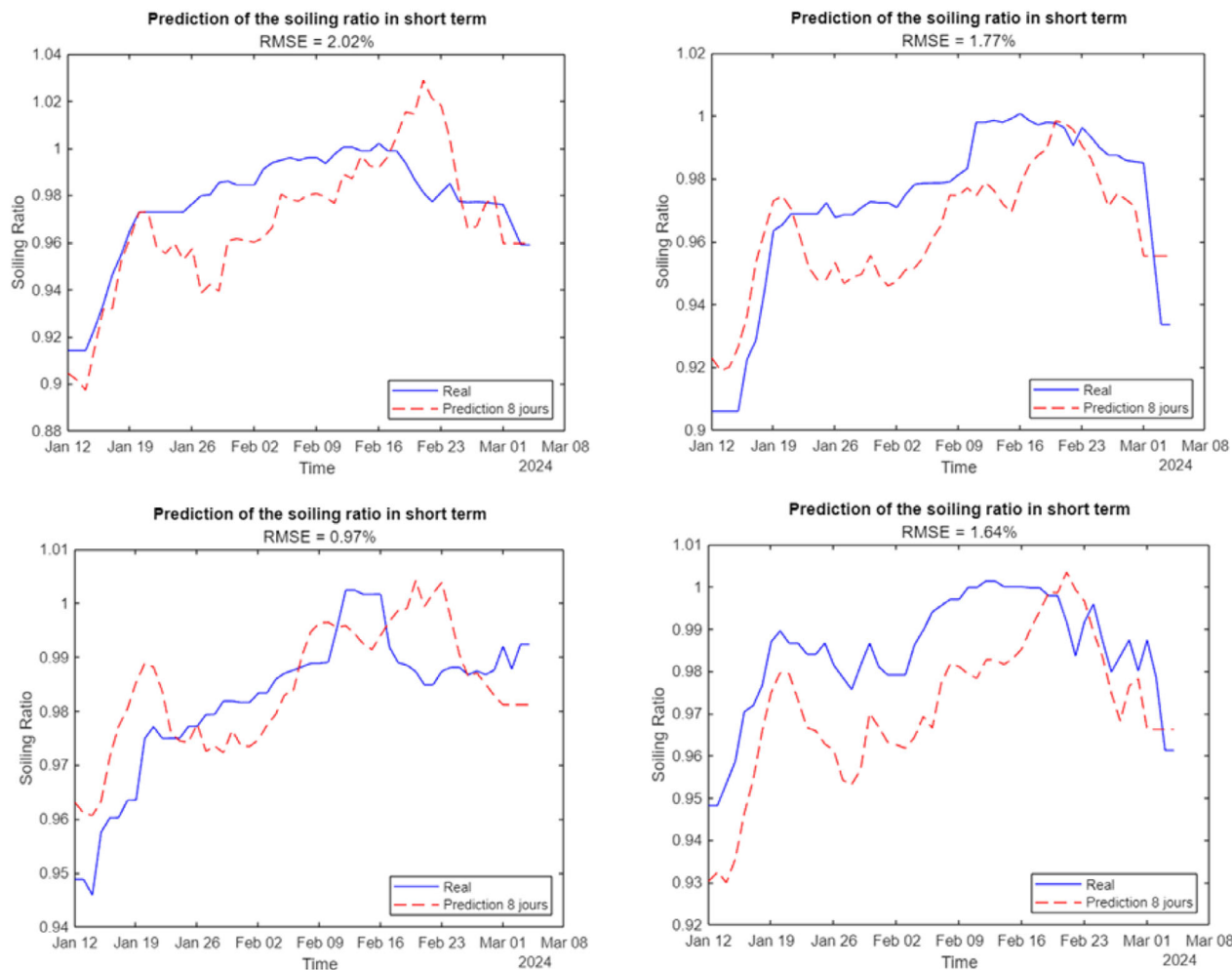


FIGURE 8 | SSQL-based predictive modeling of the SR for four different PV strings of the studied PV plant in northeastern Spain: Comparative results of predicted SR versus real SR, 8-days ahead prediction, for a study period of 3 months.

3.1 | Validation on NREL Dataset

This open-source dataset from NREL features a 1 kW monofacial silicon module system in Denver, Colorado (south-oriented, 40° tilt). DC power data from February 2010 to December 2016, along with environmental factors from a nearby weather station, were used. The model was trained on 5 years of data and tested on 2 years (Figure 5). While capable of predicting seasonal variations, the accuracy is currently low, and the predicted SR appears overly smoothed.

3.2 | Validation on a Floating PV Plant Dataset (CEA, Southern France)

This dataset originates from a floating PV plant with three zones, each having 19 junction boxes. Monofacial modules are south-oriented and tilted at 10°. DC current and power have been recorded since May 2022, complemented by environmental factors from an on-site weather station. The model, trained over 1 year and tested over 5 months (Figure 6), shows improved accuracy but requires longer-term validation.

Further simulations, employing both our SSQL method and our ML modeling framework, were carried out for the case of a utility-scale PV plant located in the Aragon region, northeastern Spain. The studied PV installation, with installed capacity of 62 MW_p, spans over 100 ha in a soiling prone site, influenced by climatic conditions (cold semi-arid climate (BSk)) that favor dust accumulations. In the specific case, for employing the SSQL approach, we examined DC current measurements at PV strings level, for a full year, identifying different SPs based on performance output (PV production) patterns. Different PV string locations experience distinct soiling behaviors due to varying microclimatic exposure conditions. Figure 7 presents indicative examples of resulting soiling profiles, generated through distinct “training,” “validation,” and “test” data. The modeling results are suggestive of considerable variations across soiling profiles for different strings, highlighting the location-dependency and influence of microclimatic conditions on soiling losses, particularly in the case of wide-area soiling-prone PV installations.

Although the SSQL method does not rely on weather data as inputs for its computations, environmental parameters such as wind speed, humidity, and precipitation were analyzed to validate

and interpret the soiling trends observed in the electrical data. This auxiliary analysis helps confirm whether environmental conditions align with SSQL-detected cleaning and soiling events. For instance, the correlation between increased humidity and dust adhesion, or the impact of heavy rainfall on natural cleaning, provides insights into the accuracy of SSQL-based estimations.

Finally, Figure 8 presents example comparative results on the SR predicted (8 days ahead) by applying the SSQL method against actual SR measurements, for an indicative 3-month period in 2024 across four different PV strings of the studied plant. Prediction errors range between 1% and 2.6%.

Although the horizon of Figure 8 appears short, it was selected as a proof-of-concept demonstration of SSQL's predictive capability under real operational conditions, based on the available monitoring window. The results are encouraging: while this limited measurement period does not yet allow the model to capture long-term seasonal tendencies, the low observed errors indicate that SSQL already provides reliable short-term predictions of soiling losses

in real-life cases. To complement this, longer-term validations have already been carried out with other datasets, including 7 years of NREL data (Denver, USA) and 1.5 years from a CEA partner's floating PV plant (Southern France), as shown earlier, in Figure 5 and 6. These extended analyses demonstrate that the ML-based component is able to capture seasonal variations, whereas SSQL remains primarily a retrospective diagnostic tool, designed for robust short-term predictions. Ongoing work, including datasets from Atacama Desert plants, will further extend the validation of SSQL over longer horizons by conditioning the methodology with seasonal priors.

To contextualize the performance of the presented SSQL and ML methodologies, Table 1 provides a comparative overview with widely used soiling models reported in the literature. While direct one-to-one benchmarking was not feasible due to differences in available datasets, monitoring infrastructure, and climatic conditions, the table synthesizes typical input requirements, core assumptions, reported error ranges, and strengths/limitations of representative models. This comparison allows situating the

TABLE 1 | Comparative overview of existing soiling loss modeling/prediction methods and the proposed SSQL/ML hybrid approach.

Model / approach	Inputs required	Core assumptions	Reported accuracy (Error range)	Strengths (✓)Limitations (✗)
Kimber model [10]	Rainfall (thresholds), dry period length	Linear accumulation; complete cleaning when rainfall exceeds threshold	≈3–5% (typical)	✓ Simple, low data requirements ✗Over-simplifies partial cleaning; ignores PM concentration; poor generalization
HSU model [11]	PM10/PM2.5 concentration, deposition velocity, tilt angle	Deposition-driven accumulation	≈1.5–3% (reported)	✓Accounts for particle size and atmospheric loading ✗Requires high-quality air quality data; site-specific
SOMOSclean [13, 25]	Rainfall (intensity/accumulation), dust events	Exponential soiling saturation; variable cleaning thresholds	≈0.7% (best reported)	✓High accuracy; dynamic handling of rainfall ✗Requires calibration; data-intensive
SRR method [15]	PV yield data only	Automated detection of cleaning and soiling events	1–3% (literature)	✓No external sensors; robust in data-only contexts ✗May misclassify events in highly variable climates
Proposed SSQL	PV PerfMs only (Idc, Pdc, PRdc, Pac)	Statistical segmentation of cleaning, stable, soiling phases	1–2.6% (Spanish case, 8-day horizon)	✓Independent of meteorological inputs; probabilistic uncertainty quantification ✗Does not yet capture seasonal cycles; retrospective focus
Proposed ML-based model	Environmental (T, RH, wind, rainfall, PM), system properties	Data-driven regression/ANNs, trained on historical datasets	2–3% (NREL, CEA floating PV cases)	✓Captures seasonal variations; scalable to diverse climates ✗Requires large datasets; risk of overfitting to site
Proposed hybrid (SSQL + ML)	Combination of electrical performance + environmental data	Retrospective + predictive integration	Comparable to SOMOSclean/SRR (1–3%)	✓Robust across data-scarce and data-rich environments; predictive capability ✗Needs longer-term validation in multiple climates

proposed approaches within the broader landscape of soiling research, highlighting both their originality and their complementarity with existing methods.

The SQL method achieves prediction errors (1–2.6%) that are comparable to more data-intensive approaches such as HSU and SOMOSclean, despite requiring no environmental or meteorological inputs. This makes it particularly attractive for sites where access to high-quality environmental data is limited, but where electrical performance data are routinely collected. Meanwhile, the ML-based model demonstrates seasonal sensitivity at error levels similar to SRR and other advanced data-driven methods, benefiting from its ability to integrate multiyear environmental datasets and capture complex, nonlinear interactions.

Taken together, the hybrid SQL + ML framework offers a balanced solution: it is robust across both data-scarce and data-rich environments, capable of short-term retrospective diagnosis as well as forward-looking seasonal prediction. Rather than aiming to replace existing models, it complements them by bridging the gap between simple empirical approaches and highly data-intensive methods. This flexibility positions the framework as a practical tool for both research applications and operational decision-making in PV plant management.

Next to the presented predictive modeling framework for PV soiling losses, CEA has also developed a virtual case study generator to support further soiling model development. This tool simulates daily soiling rates based on multivariable regression models, considering wind speed, rainfall, and airborne dust concentration. It generates synthetic datasets useful for validating and improving soiling models under controlled conditions. The ability to predict daily SRs provides valuable insights for optimizing maintenance strategies.

Further investigation reveals that soiling losses exhibit nonlinear degradation patterns, influenced by meteorological anomalies. Periods of high relative humidity combined with low wind speeds create conditions that promote dust adhesion, significantly increasing soiling rates. Conversely, heavy precipitation events can lead to unexpected recovery in performance.

Overall results of this work also highlight discrepancies between pyranometer-based soiling estimations and electrical-based soiling quantifications. While irradiance sensors provide a useful indicator of soiling trends, their readings can be affected by sensor defects or degradation, spectral variations, and angular dependencies. By cross-validating multiple data sources, the robustness of soiling assessments can be significantly improved. To further refine soiling predictions, the integration of soiling test kit-based monitoring data – such as those acquired by CEA's SoilRatio presented in [7] – is explored. Deploying dedicated soiling sensors alongside electrical performance monitoring enhances the capability to detect early-stage soiling events. Real-time data assimilation into ML models improves the responsiveness of predictive tools, enabling operators to schedule cleaning interventions more efficiently.

By integrating SQL and ML-based methodologies into PV operational management, plant operators gain a reliable framework for optimizing cleaning schedules and mitigating long-term

energy losses. These methods contribute to better economic efficiency and strategic planning for PV asset management. The insights gained from this study pave the way for integrating predictive soiling assessments into PV monitoring platforms, providing long-term benefits for system performance optimization.

4 | Conclusions – Future Work

The field of PV soiling modeling and prediction is advancing quickly due to the critical need to reduce energy losses and improve operational efficiency. This study addresses this need by developing two complementary methodologies: the SQL method and a ML-based predictive model. These approaches are designed to be integrated into PV monitoring platforms to optimize maintenance strategies and minimize energy yield losses.

The SQL method provides a robust tool for retrospective analysis and quantification of soiling impact using only electrical performance data. Unlike many traditional methods, it does not require external meteorological inputs. By analyzing PV performance trends, the SQL method identifies and classifies key soiling phases, including cleaning periods, St.Ps, and SPs. Through Monte Carlo simulations, it generates probable soiling profiles, enabling the estimation of average SRs and uncertainty ranges. This probabilistic approach helps define different soiling scenarios, from worst-case to optimal conditions, and provides valuable insights for optimizing maintenance strategies. The method also allows for the classification of different soiling accumulation patterns, distinguishing between slow, gradual accumulation, and rapid degradation. Preliminary results validate the robustness of the SQL method, with generated profiles aligning closely with observed soiling trends. Field studies at a utility-scale PV plant in Spain demonstrated that the method could predict SRs 8 days ahead with prediction errors ranging from 1% to 2.6%. While the current SQL tool is primarily for retrospective analysis, the next phase of development will focus on making it a predictive tool by conditioning Monte Carlo simulations with weather forecasts and seasonal priors. This will enable a more realistic representation of seasonal variability in soiling profiles and further improve the reliability of cleaning optimization strategies.

The ML-based model complements SQL by providing a forward-looking predictive capability using a comprehensive set of environmental parameters. This approach integrates ANNs and regression models, applying data pre-processing techniques to enhance accuracy. By utilizing hourly data and training the models over extended periods, this method is adept at capturing seasonal variations and complex, nonlinear relationships that traditional models might miss. Validation studies using datasets from NREL and a CEA floating PV plant demonstrate the model's ability to predict seasonal soiling variations. These validations show that longer training periods improve predictive accuracy. The model also incorporates regional climate variations and site-specific soiling characteristics to improve its predictive performance.

Despite these advancements, the study highlights several remaining challenges in soiling modeling. Many ML models are limited

in their generalizability because they are trained and tested on data from the same system or location. Additionally, powerful ML models require large datasets, which can limit their applicability in regions with sparse monitoring. The study also notes discrepancies between pyranometer-based soiling estimations and electrical-based soiling quantifications, and suggests that cross-validating multiple data sources can significantly improve the robustness of soiling assessments. To address these issues, future work will focus on improving model generalizability across diverse environments, refining input parameters, and exploring advanced ML techniques such as time-series neural networks and recurrent neural networks. The model will also be adapted for bifacial PV systems and validated against direct measurements from a dedicated soiling sensor developed by CEA. The development of a virtual case study generator further supports this effort by providing diverse synthetic datasets for model validation and improvement under controlled conditions. By integrating these two powerful methodologies, this work provides a reliable framework for PV plant operators to make data-driven decisions that lead to better economic efficiency and strategic planning.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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